

Area Between Curves

ANSWER KEY



Area between a curve and the x-axis

- 1 Find the area of the region bounded by $y = 4 - x^2$ and the x -axis.

Roots at $x = \pm 2$. Area $= \int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

Area between two curves (vertical strips)

- 2 Find the area between $y = x^2$ and $y = x + 2$.

Intersect: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$: $x = -1, 2$. On this interval $x + 2 \geq x^2$.
Area $= \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} - \left(-\frac{7}{6} \right) = \frac{9}{2}$.

- 3 Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.

On $[0, 1]$, $\sqrt{x} \geq x^2$. Area $= \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3} x^{3/2} - \frac{x^3}{3} \right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$.

Regions requiring horizontal strips

- 4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y .

Intersect: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0$: $y = -1, 2$. On this interval $y + 2 \geq y^2$.
Area $= \int_{-1}^2 [(y + 2) - y^2] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{9}{2}$.

Regions with multiple intersection points

- 5 Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

On $[0, \frac{\pi}{4}]$, $\cos x \geq \sin x$; on $[\frac{\pi}{4}, \pi]$, $\sin x \geq \cos x$. Total area
 $= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi} (\sin x - \cos x) dx$. First: $[\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1$. Second:
 $[-\cos x - \sin x]_{\pi/4}^{\pi} = (1 - 0) - (-\sqrt{2}) = 1 + \sqrt{2}$. Total $= (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2} \approx 2.83$.

Area using the more efficient variable of integration

- 6 Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.

Solving for x : $x = 4 - y^2$, bounded by $x = 0$ (i.e. $y = \pm 2$). Integrating with respect to y is easier: Area
 $= \int_{-2}^2 (4 - y^2) dy = \left[4y - \frac{y^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

Setting up but not evaluating (AP-style)

- 7 Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.
On $[1, 2]$, $e^x \geq \ln x$ (since e^x grows much faster). Area $= \int_1^2 (e^x - \ln x) dx$.
- 8 Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x -axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.
On $[-2, 0]$, $x^3 - 4x \geq 0$; on $[0, 2]$, $x^3 - 4x \leq 0$. Total area
 $= \int_{-2}^0 (x^3 - 4x) dx + \int_0^2 -(x^3 - 4x) dx = \int_{-2}^0 (x^3 - 4x) dx - \int_0^2 (x^3 - 4x) dx$.

Area between curves and a table of values

- 9 Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)] dx$.
Let $h(x) = f(x) - g(x)$: $h(0) = 7$, $h(2) = 4$, $h(4) = 4$, $h(6) = 6$. Trapezoidal Rule with width 2:
 $T = 2 \left[\frac{7+4}{2} + \frac{4+4}{2} + \frac{4+6}{2} \right] = 2[5.5 + 4 + 5] = 2(14.5) = 29$.

Visualizing area between two curves

- 10 The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.
The line $y = x + 2$ lies above the parabola $y = x^2$ throughout $(-1, 2)$, matching the earlier setup
 $\int_{-1}^2 [(x + 2) - x^2] dx = \frac{9}{2}$.

A capstone area problem

- 11 Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.
On $[-1, 0]$, $x^3 - x \geq 0$; on $[0, 1]$, $x^3 - x \leq 0$. Area $= \int_{-1}^0 (x^3 - x) dx + \left| \int_0^1 (x^3 - x) dx \right|$.
 $\int_{-1}^0 (x^3 - x) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 = 0 - \left(\frac{1}{4} - \frac{1}{2} \right) = \frac{1}{4}$. By symmetry the second piece is also $\frac{1}{4}$. Total area $= \frac{1}{2}$.

Area between an exponential curve and a line

- 12 Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x , tangent only at $x = 0$).
Area $= \int_0^2 [e^x - (x + 1)] dx = \left[e^x - \frac{x^2}{2} - x \right]_0^2 = (e^2 - 2 - 2) - (1 - 0 - 0) = e^2 - 5$.

An applied area-between-curves problem

- 13** Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

Intersection: $50 - 2t = 30 + t \Rightarrow 20 = 3t \Rightarrow t = \frac{20}{3} \approx 6.67$. For $t < \frac{20}{3}$, $P_1 > P_2$; for $t > \frac{20}{3}$, $P_2 > P_1$. Total absolute difference = $\left| \int_0^{20/3} (P_1 - P_2) dt \right| + \left| \int_{20/3}^{10} (P_2 - P_1) dt \right|$, which can be computed piecewise using $P_1 - P_2 = 20 - 3t$.